

uses of optical fibres in medicine

light travels down optical fibres into or out of body:

TO examine internal organ

light travels both ways into and out of body

TO destroy (cancerous cells)

by heating them

OR used in endoscope (bundle of fibres)

which is put into body to view internal organ OR for keyhole surgery

conditions for TIR

incident ray from optically dense medium

angle of incidence greater than critical angle

no light refracted

reflected with $i=r$

describe what is meant by transverse wave motion

transmission of energy (through medium) AND no transfer of energy;

direction of vibration of particles;

perpendicular to direction to energy travel/wave/propagation;

slowing of light in terms of wavefronts

one side of wavefront slows down before other side;

wavefront slews around OR bends at boundary;

bends towards the normal OR bends towards the side that slows first;

why X-rays okay for patient but dentist needs to be shielded

patient rarely exposed

low total dose on patient

meaningful comment about benefit outweighs danger

dentist frequently exposed

total dose on dentist will be high if not shielded

optical fibres in communication tech

information/message/sound/signal/data encoded as pulses of light is sent through fibre;

light (or IR radiation) travels along fibre;

light detected at far end or message decoded or total internal reflection mentioned;

what happens when incident ray increases the angle of incidence to critical angle

angle (from normal) of refraction increases

refracted ray travels along boundary OR reflected ray becomes brighter

critical angle reached

light reflects back into glass with $i = r$

use of optical fibres

to carry telephone signals;

medical diagnosis/imaging;

specified artistic display;
specified lighting;